Steps 1a–1b

- Generic Gaussian probabilistic inference solution can be divided into two main steps, each having three sub-steps.
- **General step 1a**: State prediction time update.
  - Each time step, compute an updated prediction of the present value of x_k based on prior information and the system model:

$$\hat{x}_k^- = \mathbb{E}\big[x_k \mid \mathbb{Z}_{k-1}\big] = \mathbb{E}\big[f(x_{k-1}, u_{k-1}, w_{k-1}) \mid \mathbb{Z}_{k-1}\big].$$

- **General step 1b**: Prediction-error covariance time update.
  - Compute the prediction-error covariance matrix $\Sigma_{\tilde{x},k}^-$ based on prior information and the system model.
  - We compute $\Sigma_{\tilde{x},k}^- = \mathbb{E}\big[(\tilde{x}_k^-)(\tilde{x}_k^-)^T\big]$, where $\tilde{x}_k^- = x_k - \hat{x}_k^-$.

Step 1c

- **General step 1c**: Predict system output.
  - Predict the system's output using prior information:

$$\hat{z}_k = \mathbb{E}\big[z_k \mid \mathbb{Z}_{k-1}\big] = \mathbb{E}\big[h(x_k, u_k, v_k) \mid \mathbb{Z}_{k-1}\big].$$

- Summarizing general step 1:

Step 1a: State prediction time update

$$\hat{x}_k^- = \mathbb{E}\big[x_k \mid \mathbb{Z}_{k-1}\big] = \mathbb{E}\big[f(x_{k-1}, u_{k-1}, w_{k-1}) \mid \mathbb{Z}_{k-1}\big].$$

Step 1b: Prediction-error covariance time update

$$\Sigma_{\tilde{x},k}^- = \mathbb{E}\big[(\tilde{x}_k^-)(\tilde{x}_k^-)^T\big] = \mathbb{E}\big[(x_k - \hat{x}_k^-)(x_k - \hat{x}_k^-)^T\big].$$

Step 1c: Predict system output

$$\hat{z}_k = \mathbb{E}\big[z_k \mid \mathbb{Z}_{k-1}\big] = \mathbb{E}\big[h(x_k, u_k, v_k) \mid \mathbb{Z}_{k-1}\big].$$

Prediction

Step 2

- **General step 2a**: Estimator gain matrix.
  - Compute the estimator gain matrix $L_k = \Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1}$.
- **General step 2b**: State estimate measurement update.
  - Compute the state estimate by updating the prediction using the L_k and the innovation $z_k - \hat{z}_k$:

$$\hat{x}_k^+ = \hat{x}_k^- + L_k(z_k - \hat{z}_k).$$

- **General step 2c**: Estimation-error covariance measurement update.
  - Compute the estimation-error covariance matrix:

$$\begin{aligned}\Sigma_{\tilde{x},k}^+ &= \mathbb{E}\big[(\tilde{x}_k^+)(\tilde{x}_k^+)^T\big]\\ &= \Sigma_{\tilde{x},k}^- - L_k \Sigma_{\tilde{z},k} L_k^T.\end{aligned}$$

Summarizing step 2

- Summarizing general step 2:

Step 2a: Estimator gain matrix
$$L_k = \Sigma^-_{\tilde{x}\tilde{z},k}\Sigma^{-1}_{\tilde{z},k}.$$
Step 2b: State estimate measurement update
$$\hat{x}^+_k = \hat{x}^-_k + L_k(z_k - \hat{z}_k).$$
Step 2c: Estimation-error covariance measurement update
$$\Sigma^+_{\tilde{x},k} = \Sigma^-_{\tilde{x},k} - L_k\Sigma_{\tilde{z},k}L_k^T.$$

Correction

- **KEY POINT:** Estimator outputs state estimate $\hat{x}^+_k$, error covariance estimate $\Sigma^+_{\tilde{x},k}$.
  - That is, we have high confidence that the truth lies within $\hat{x}^+_k \pm 3\sqrt{\operatorname{diag}(\Sigma^+_{\tilde{x},k})}$.
  - Estimator then waits until next sample interval, updates k, proceeds to step 1a.

Summary

Step 1a: State prediction time update
$$\hat{x}^-_k = \mathbb{E}\big[x_k \mid \mathbb{Z}_{k-1}\big] = \mathbb{E}\big[f(x_{k-1}, u_{k-1}, w_{k-1}) \mid \mathbb{Z}_{k-1}\big].$$
Step 1b: Prediction-error covariance time update
$$\Sigma^-_{\tilde{x},k} = \mathbb{E}\big[(\tilde{x}^-_k)(\tilde{x}^-_k)^T\big] = \mathbb{E}\big[(x_k - \hat{x}^-_k)(x_k - \hat{x}^-_k)^T\big].$$
Step 1c: Predict system output
$$\hat{z}_k = \mathbb{E}\big[z_k \mid \mathbb{Z}_{k-1}\big] = \mathbb{E}\big[h(x_k, u_k, v_k) \mid \mathbb{Z}_{k-1}\big].$$

Prediction

Step 2a: Estimator gain matrix
$$L_k = \Sigma^-_{\tilde{x}\tilde{z},k}\Sigma^{-1}_{\tilde{z},k}.$$
Step 2b: State estimate measurement update
$$\hat{x}^+_k = \hat{x}^-_k + L_k(z_k - \hat{z}_k).$$
Step 2c: Estimation-error covariance measurement update
$$\Sigma^+_{\tilde{x},k} = \Sigma^-_{\tilde{x},k} - L_k\Sigma_{\tilde{z},k}L_k^T.$$

Correction